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$$-i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 = 0 \\ 2xy = -1 \end{cases} \quad \begin{cases} x^2 - y^2 = 0 \\ xy = \frac{-1}{2} \end{cases} \quad \begin{cases} x^2 - y^2 = 0 \\ x^2(-y)^2 = \frac{1}{4} \end{cases}$$

$$\lambda^2 - 0\lambda - \frac{1}{4} = 0$$

$$\Delta = -4 \cdot \left(\frac{-1}{4}\right) = 1$$

$$\lambda_1 = \frac{0+1}{2} = \frac{1}{2}$$

$$\lambda_2 = \frac{0-1}{2} = \frac{-1}{2}$$

$$x = \pm \frac{\sqrt{2}}{2}, y = \pm \frac{\sqrt{2}}{2}$$

$$x + yi \in \left\{ \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i, -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right\}$$

References